



Cloud's roles in advancing data analytics

Keeping track of business data used to be incredibly labor-intensive.

From handwritten ledgers to complex filing systems to typing facts and figures into a spreadsheet.

Collecting, cleaning, organizing, and storing information was a huge and resource-heavy task.

But cloud data analytics has automated and enhanced these processes, making data management much more efficient and less prone to human error.

In this video, we'll dive into how the cloud enables data from a variety of sources to be smoothly integrated, creating a single source that users can access and analyze in real time.

First, cloud data can be managed with data integration or data ingestion.

Data integration combines data from different sources into a single, usable data source.

This integration can happen through the ETL process, or extract, transform, and load; or the ELT process: extract, load, and transform.

ETL and ELT are cloud-based approaches that use the power of cloud data warehouses like Google BigQuery to transform data.

ETL transforms the data before it's loaded into the warehouse and ELT transforms it after.

But either way, it's ready for further processing or analysis.

Data ingestion obtains, imports, and processes data for later use or storage.

The information is obtained from various sources and processed through stream or batch ingestion.

Stream ingestion involves real-time continuous processing of data as soon as it is collected from various sources.

Batch ingestion processes data in predefined intervals or larger chunks.

Those are just a few of the ways cloud data analytics has transformed how organizations access their data.

There are also web interfaces, application programming interfaces or APIs, SQL, other ingestion tools like Pub/Sub, and business intelligence solutions, like Looker and Jupyter Notebooks.

All of these help users access data that's stored in the cloud anywhere, anytime.

And while we're discussing data in the cloud, cloud data analytics also makes it possible to store different types of data like files, objects, or blocks.

File data is information that's stored in a file on your computer or another storage device.

Object data is a piece of information with a unique identifier which you can find no matter where it's stored.

And block data is a piece of information that has been cut from a larger piece of information and given its own file path.

So many data analysis activities have greatly benefited from cloud data analytics processes: big-data analysis, the ability

for visualization of multiple data sources asynchronously, AI and ML, custom report analysis, data mining, data science.

The list goes on and on!

In today's world, innovation is the driving force for many companies.

And there's no doubt that data fuels these innovations.

It's really amazing how much cloud analytics has advanced the field of data analytics, making powerful analytical tools and processes available to organizations of all kinds.

At the same time, it enhances the analytics process, making it easier, faster, and more cost-effective for users to discover valuable insights from data.